DAYS 36 TO 40 | CAPSTONE

Capstone Guide

Five days, one real-world product, built end to end by your team. Tue 15 Dec to Mon 21 Dec 2026.

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1. What the capstone is

For the final week you work in teams of 3 to 4 and build a complete product for a realistic customer scenario, using everything from Days 1 to 35: Python, APIs, databases, React, testing, data pipelines, cloud and containers, GenAI, agents, security, evaluation and LLMOps. You also practise the FDE Mindset skills: scoping with a customer, handling ambiguity, communicating progress and presenting results.

Goal: a working, tested, deployable product with measured quality, and a confident ten minute demo. Depth beats breadth: a smaller scope done well scores higher than many half-finished features.

2. Choose one of six projects

#	Project	Core skills
01	ShopSmart: E-commerce Recommendation and Shopping Assistant	Embeddings for product descriptions; hybrid search with RRF
02	BookFinder: Library Book Finder with Natural Language Search	Embeddings and hybrid search with RRF
03	PolicyPal: Enterprise HR and IT Policy Chatbot	Context engineering and chunking
04	SupportPilot: Agentic Customer Support Assistant	Tool calling and LangGraph
05	InsightDesk: Retail Sales Pipeline and Natural Language Data Agent	Data agent with sandboxed code
06	LegacyLift: Modernise a Legacy Order-Management App (NEW v4)	Code modernisation with OpenAI Codex, tests first

Full briefs are in the Projects folder. Each team picks one project in the first hour of Day 36. Two teams may choose the same project; they will be compared on the same checklist.

3. Technical baseline (every project)

- FastAPI back end with Pydantic models and at least 6 endpoints (Days 5 to 7)
- A database: SQL (SQLAlchemy) and/or MongoDB (Days 4, 6, 7)
- React front end with loading, error and empty states (Days 10, 11)
- An AI component from the GenAI days: RAG, recommendation with embeddings, agent or data agent (Days 12 to 28)
- Tests with PyTest and a GitHub or Azure DevOps CI pipeline (Days 15, 17, 20)
- Dockerfile and docker-compose; Azure deployment or documented steps (Days 20, 21, 34)
- Guardrails and secrets handling (Days 7, 30)

- Golden set, metrics and a quality gate (Days 31, 32)
- Structured logs and a monitoring report or dashboard (Day 35)

4. Baseline additions in v4 (every project) NEW (v4)

- **One modernisation step:** pick one legacy module named in your brief, write characterisation tests, then migrate it in small verified steps with OpenAI Codex (Days 23, 30).
- **AGENTS.md in the team repo:** build, test and style commands, folders not to touch, and "never put secrets in prompts" (Day 3, template in shared/templates/AGENTS_learner_template.md).
- **Security checklist** mapped to the OWASP Top 10 for Agentic Applications (shared/templates/OWASP_Agentic_Checklist.md; Days 23, 30).
- **Vulnerability report** from bandit, pip-audit, npm audit and your red-team suite (shared/templates/Vulnerability_Report_Template.md; Days 7, 20, 30).
- **Token and cost report:** tokens and estimated cost per request and per 1,000 requests, before and after one optimisation, with placeholder prices labelled (Days 12, 25, 33).

5. Day-by-day plan

Day 36 (Tue 15 Dec 2026): Kick-off and design

- Quiz on Day 35
- Teams of 3 to 4 formed; project chosen
- Problem statement, users, success metrics, assumptions log
- Architecture diagram and data model
- Repository, board and CI skeleton created, with AGENTS.md in the repo root (NEW v4)
- Pick your modernisation step and write its characterisation tests first (NEW v4)
- Checkpoint 1 (end of day): design review with mentor

Day 37 (Wed 16 Dec 2026): Core build

- Stand-up (15 min)
- Data loaded; back end API with main endpoints
- AI core working end to end (retrieval, recommendation, agent or pipeline)
- Golden set draft of 20 items
- Token and cost logging on every model call (NEW v4)
- Checkpoint 2: walking skeleton demo

Day 38 (Thu 17 Dec 2026): Complete features and quality

- Stand-up
- Front end connected to the API
- Guardrails and error handling
- Evaluation run and first error analysis
- Tests in CI; quality gate added
- Modernisation step merged through a reviewed pull request (NEW v4)
- Checkpoint 3: feature-complete MVP

Day 39 (Fri 18 Dec 2026): Harden, deploy, observe

- Stand-up

- Docker images; deploy to Azure or a documented local deployment
- Structured logs and a monitoring view
- Red-team and slice checks; fix top issues
- Security checklist (OWASP Top 10 for Agentic Applications) and vulnerability report completed (NEW v4)
- Token and cost report written (NEW v4)
- Demo rehearsal with peer feedback
- Code freeze at 17:00 IST

Day 40 (Mon 21 Dec 2026): Demo day

- Deliverables due at 09:00 IST
- Final rehearsal (morning)
- Demo: 10 minutes + 5 minutes Q&A per team
- Evaluator scoring with the rubric
- Individual reflection and peer feedback
- Closing and certificates

6. Daily stand-up (15 minutes)

Each person answers three questions: what I finished yesterday, what I will finish today, what is blocking me. The team updates its board and decision log. Mentors join at least once a day. In the evening, follow Hotel_Lab.md to prepare the next stand-up.

7. Rules

- OpenAI Codex (VS Code extension, program API key, approved models only) is the AI coding tool. Work through reviewed pull requests. Disclose in the README which parts were generated and how you reviewed them.
- Never paste API keys, customer data or credentials into prompts or AGENTS.md. Check daily token usage.
- Use only the generated or public sample data. Never use real customer or personal data.
- Keep secrets out of Git. Use environment variables or a secret store such as Azure Key Vault.
- Every team member must be able to explain every part of the system during Q&A.
- Cloud resources (Azure window 3, Days 38 to 40): use tags, stay within the agreed budget, and delete resource groups after demo day.

8. Deliverables (due Day 40, Mon 21 Dec 2026, 09:00 IST)

Deliverable	Details
Repository	Code, tests, Dockerfiles, pipeline YAML, golden set, AGENTS.md (NEW v4), README with setup, architecture and AI tool disclosure
Architecture diagram	Components, data flow, where AI is used, guardrails
Evaluation report	Golden set results, metrics, error analysis, slices, known limits
Operations note	How to deploy, monitor, roll back; one runbook
Security checklist (NEW v4)	OWASP Top 10 for Agentic Applications, one line of evidence per item

Deliverable	Details
Vulnerability report (NEW v4)	Findings, severity, fix, verified, from scans and red-team tests
Token and cost report (NEW v4)	Tokens and estimated cost per request and per 1,000 requests, before and after one optimisation (placeholder prices labelled)
Modernisation note (NEW v4)	The legacy module, characterisation tests, the steps taken and the PR links
Demo deck	8 to 12 slides: problem, solution, demo, results, limits, next steps
Decision log	Key decisions with options, reasons and dates

9. Demo day format

- 10 minutes: problem and users (1), live demo (5), results and evaluation (2), limits and next steps (2).
- 5 minutes of Q&A from evaluators acting as the customer's sponsor, IT lead and security lead.
- Every team member speaks. Lead with the answer, show evidence, be honest about limits.

10. Evaluation rubric (100 points)

Criterion	Points	What evaluators look for
Problem understanding and scoping	10	Clear problem statement, users, success metrics, assumptions log, agreed MVP scope (FDE Mindset Days 1, 14, 19).
Architecture and design	10	Sensible architecture diagram, layered code, right tool choices with reasons, data model (Days 4, 6, 18, 34).
Core functionality	20	All must-have features work end to end in the demo; project checklist items delivered.
AI and data quality	15	Golden set of at least 20 items, measured metrics (retrieval, faithfulness or recommendation quality), error analysis (Days 22, 31, 32).
Engineering quality	10	Readable code, tests (API, pipeline, prompts), Git history with pull requests, CI pipeline green (Days 15, 17, 20). NEW (v4): AGENTS.md in the repo; modernisation step done tests first.
Security, guardrails and responsible AI	10	Input and output guardrails, secrets handling, least privilege, a red-team check, fairness or language slice check (Days 7, 30). NEW (v4): OWASP Agentic checklist and vulnerability report.
Deployment and operations	10	Containerised, deployable (Azure or documented), structured logs, a dashboard or report, one drift or quality check (Days 21, 33 to 35). NEW (v4): token and cost report.
Demo and communication	10	Answer-first readout, live demo, honest limits, confident Q&A, clear handover README (FDE Mindset Days 24, 29).
Teamwork and process	5	Daily stand-ups, decision log, fair contribution, kind and specific feedback (FDE Mindset Days 14, 24, 29).

Each criterion is scored on four levels; points = weight x level / 4.

Level	Meaning
4 Excellent	Goes beyond the brief; production-minded; evidence for every claim.
3 Good	Meets the brief fully with minor gaps.
2 Developing	Partly meets the brief; important gaps or untested parts.
1 Beginning	Major parts missing or not working.

Grade bands: 85 to 100 Distinction, 70 to 84 Merit, 55 to 69 Pass, below 55 Needs resubmission. Individual scores can be adjusted by up to 10 points based on peer feedback and Q&A.

11. Submission checklist

- README lets a stranger run the project in under 15 minutes
- CI pipeline is green on main, with security gates (bandit, pip-audit, npm audit)
- Golden set and evaluation report are committed
- AGENTS.md, security checklist, vulnerability report and token and cost report are committed (NEW v4)
- No secrets or personal data in the repository
- Demo works without internet surprises (have a recorded backup)
- Azure resources tagged and scheduled for deletion

Ganbatte (頑張って): do your best